

## CO3 AT1 - CASE STUDY

### Case study 1:

**Q1. Calculate the customer's z-score.**

**Answer**

The customer's **Z-score is 3**.

**Given**

- Mean ( $\mu$ ) = ₹2000
- Standard Deviation ( $\sigma$ ) = ₹400
- Customer Spending ( $X$ ) = ₹3200

**Formula**

$$Z = \frac{X - \mu}{\sigma}$$

**Calculation**

$$Z = \frac{3200 - 2000}{400} = \frac{1200}{400} = 3$$

**Steps**

1. Write the given values.
2. Apply the Z-score formula.
3. Substitute the values into the formula.
4. Perform the calculation.
5. Write the final answer.

**Final Answer**

**Customer's Z-score = 3**

**Q2. Is the customer's spending unusually high compared with the average?**

**Answer**

**Yes.** The customer's spending is unusually high because the Z-score is 3.

**Interpretation Table**

| Z-score | Interpretation |
|---------|----------------|
|---------|----------------|

|                    |                |
|--------------------|----------------|
| $Z > 2$            | Unusually High |
| $-2 \leq Z \leq 2$ | Normal         |
| $Z < -2$           | Unusually Low  |

#### Steps

1. Take the calculated Z-score.
2. Compare it with the interpretation table.
3. Since  $3 > 2$ , the spending is unusually high.
4. Write the conclusion.

#### Final Answer

**The customer's spending is unusually high.**

#### Q3. What would a negative z-score mean in this context?

#### Answer

A negative Z-score indicates that the customer spends **below the average amount**.

#### Interpretation Table

| Z-score      | Meaning          |
|--------------|------------------|
| Positive (+) | Above Average    |
| Zero (0)     | Equal to Average |
| Negative (-) | Below Average    |

#### Steps

1. Observe the sign of the Z-score.
2. A negative sign means the value is below the mean.
3. Therefore, the customer spends less than the average.

#### Final Answer

**A negative Z-score means below-average spending.**

#### Q4. If another customer has a z-score of $-1.5$ , what does it tell you?

### Answer

The customer spends **1.5 standard deviations below the average spending.**

### Interpretation Table

| Customer   | Z-score | Interpretation          |
|------------|---------|-------------------------|
| Customer 1 | +3      | Unusually High Spending |
| Customer 2 | -1.5    | Below Average Spending  |

### Steps

1. Note the given Z-score ( $-1.5$ ).
2. A negative Z-score indicates below-average spending.
3. The value **1.5** shows the customer is **1.5 standard deviations below the mean.**
4. Write the conclusion.

### Final Answer

**The customer spends 1.5 standard deviations below the average.**

## CASE STUDY – 2: Delivery Time Analysis

### Given Data

#### Delivery Time (minutes)

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

### Q1. Calculate the Mean and Median.

#### Answer

- Mean = 32.6 minutes
- Median = 30 minutes

#### Given

Delivery Time = 25, 27, 28, 29, 30, 30, 31, 32, 34, 60

Number of observations (**n**) = 10

#### Formula

##### Mean

$$Mean = \frac{\sum X}{n}$$

##### Calculation

$$\sum X = 25 + 27 + 28 + 29 + 30 + 30 + 31 + 32 + 34 + 60 = 326 \quad Mean = \frac{326}{10} = 32.6$$

##### Median

Since **n = 10 (even number)**,

$$Median = \frac{5^{th} + 6^{th} \text{ values}}{2} = \frac{30 + 30}{2} = 30$$

#### Steps

1. Find the total of all observations.
2. Divide the total by the number of observations to get the mean.
3. Arrange the data in ascending order.
4. Find the middle two values.
5. Calculate their average to get the median.

**Final Answer**

- Mean = 32.6 minutes
- Median = 30 minutes

**Q2. Find Q1 and Q3.****Answer**

- Q1 = 28
- Q3 = 32

**Given**

Data (Ascending Order):

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

**Calculation**

Lower Half:

25, 27, 28, 29, 30

**Q1 = 28**

Upper Half:

30, 31, 32, 34, 60

**Q3 = 32**

**Steps**

1. Arrange the data in ascending order.
2. Divide the data into two halves.
3. Find the median of the lower half (Q1).
4. Find the median of the upper half (Q3).

**Final Answer**

- Q1 = 28
- Q3 = 32

**Q3. Calculate the IQR.****Answer**

$$\mathbf{IQR = 4}$$

**Given**

- $Q1 = 28$
- $Q3 = 32$

**Formula**

$$IQR = Q3 - Q1$$

**Calculation**

$$IQR = 32 - 28 = 4$$

**Steps**

1. Write Q1 and Q3.
2. Apply the IQR formula.
3. Subtract Q1 from Q3.
4. Write the result.

**Final Answer**

$$\mathbf{IQR = 4}$$

**Q4. Check for Outliers using the  $1.5 \times \text{IQR}$  Rule.**

**Answer**

**60 is an outlier.**

**Given**

- $Q1 = 28$
- $Q3 = 32$
- $IQR = 4$

**Formula**

Lower Limit

$$Q1 - 1.5(IQR)$$

Upper Limit

$$Q3 + 1.5(IQR)$$

**Calculation**

Lower Limit

$$28 - (1.5 \times 4)28 - 6 = 22$$

Upper Limit

$$32 + (1.5 \times 4)32 + 6 = 38$$

Since **60** > **38**, it is an outlier.

### Steps

1. Calculate the IQR.
2. Find the lower limit.
3. Find the upper limit.
4. Compare all observations with these limits.
5. Identify the outlier.

### Final Answer

**60 is an Outlier.**

**Q5. Explain whether the Mean or Median gives a better description of the typical delivery time.**

### Answer

**Median gives a better description of the typical delivery time because the dataset contains an outlier (60), which increases the mean. The median is not affected by outliers and better represents the central value.**

### Comparison Table

| Measure | Value | Remark                                  |
|---------|-------|-----------------------------------------|
| Mean    | 32.6  | Affected by Outlier                     |
| Median  | 30    | Better Represents Typical Delivery Time |

### Steps

1. Compare the mean and median.

2. Observe the presence of an outlier.
3. Understand that the mean is affected by extreme values.
4. Conclude that the median is a better measure.

### **Final Answer**

**Median is the better measure of the typical delivery time because it is not affected by the outlier.**

### **CASE STUDY – 3: Two Cities' Temperatures**

#### **Given Data**

**City A (°C) 28 29 30 30 31 30 32**

**City B (°C) 20 25 30 35 30 25 45**

**Q1. Calculate the mean temperature for each city.**

#### **Answer**

- **Mean of City A = 30°C**
- **Mean of City B = 30°C**

#### **Given**

City A = 28, 29, 30, 30, 31, 30, 32

City B = 20, 25, 30, 35, 30, 25, 45

Number of observations (**n**) = 7

#### **Formula**

$$Mean = \frac{\Sigma X}{n}$$

#### **Calculation**

##### **City A**

$$28 + 29 + 30 + 30 + 31 + 30 + 32 = 210 \quad Mean = \frac{210}{7} = 30^{\circ}C$$

##### **City B**

$$20 + 25 + 30 + 35 + 30 + 25 + 45 = 210 \quad Mean = \frac{210}{7} = 30^{\circ}C$$



### Steps

1. Find the sum of all temperatures.
2. Count the number of observations.
3. Apply the mean formula.
4. Calculate the mean.
5. Write the final answer.

### Final Answer

- **City A Mean = 30°C**
- **City B Mean = 30°C**

### Q2. Calculate the variance for each city.

#### Answer

- **Variance of City A  $\approx 1.43$**
- **Variance of City B  $\approx 57.14$**

#### Given

Mean of both cities = **30°C**

#### Formula

$$Variance = \frac{\sum(X-\mu)^2}{N}$$

#### Calculation

##### City A

| X  | X - 30 | (X - 30) <sup>2</sup> |
|----|--------|-----------------------|
| 28 | -2     | 4                     |
| 29 | -1     | 1                     |
| 30 | 0      | 0                     |
| 30 | 0      | 0                     |
| 31 | 1      | 1                     |
| 30 | 0      | 0                     |

|    |   |   |
|----|---|---|
| 32 | 2 | 4 |
|----|---|---|

$$\sum(X - \mu)^2 = 10 \text{ Variance} = \frac{10}{7} = 1.43$$

### City B

| X  | X - 30 | (X - 30) <sup>2</sup> |
|----|--------|-----------------------|
| 20 | -10    | 100                   |
| 25 | -5     | 25                    |
| 30 | 0      | 0                     |
| 35 | 5      | 25                    |
| 30 | 0      | 0                     |
| 25 | -5     | 25                    |
| 45 | 15     | 225                   |

$$\sum(X - \mu)^2 = 400 \text{ Variance} = \frac{400}{7} = 57.14$$

### Steps

1. Find the mean.
2. Calculate deviation (X-Mean).
3. Square each deviation.
4. Find the total.
5. Divide by N.

### Final Answer

- City A Variance = 1.43
- City B Variance = 57.14

**Q3. Which city has greater temperature variability?**

**Answer**

**City B has greater temperature variability.**

#### Comparison Table

| City   | Variance | Variability |
|--------|----------|-------------|
| City A | 1.43     | Low         |
| City B | 57.14    | High        |

#### Steps

1. Compare both variances.
2. Higher variance means higher variability.
3. City B has the larger variance.
4. Therefore, City B has greater variability.

#### Final Answer

**City B has greater temperature variability.**

#### Q4. Why can two datasets have the same mean but different standard deviations?

#### Answer

Two datasets can have the **same mean** but **different standard deviations** because the values may be spread differently around the mean. One dataset may have values close to the mean, while the other has values far from the mean.

#### Comparison Table

| Dataset | Mean | Spread      |
|---------|------|-------------|
| City A  | 30   | Low Spread  |
| City B  | 30   | High Spread |

#### Steps

1. Compare the means.
2. Observe the spread of values.
3. Greater spread results in a higher standard deviation.
4. Hence, both datasets can have the same mean but different standard deviations.

#### Final Answer

**Because the spread of values around the mean is different.**

**Q5. Which city has more predictable temperatures?**

**Answer**

**City A has more predictable temperatures.**

**Comparison Table**

| City   | Predictability   |
|--------|------------------|
| City A | More Predictable |
| City B | Less Predictable |

**Steps**

1. Compare the variances.
2. Lower variance means more consistency.
3. More consistency means better predictability.
4. Therefore, City A is more predictable.

**Final Answer**

**City A has more predictable temperatures.**

## CASE STUDY – 4: Data Center CPU Usage

### Given Data

#### CPU Usage (%)

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

### Q1. Calculate the Mean, Median and Mode.

#### Answer

- **Mean = 51.3%**
- **Median = 47%**
- **Mode = 45%**

#### Given

CPU Usage = 40, 42, 45, 45, 46, 48, 50, 52, 55, 90

Number of observations (**n**) = 10

#### Formula

##### Mean

$$Mean = \frac{\sum X}{n}$$

##### Calculation

$$40 + 42 + 45 + 45 + 46 + 48 + 50 + 52 + 55 + 90 = 513 \quad Mean = \frac{513}{10} = 51.3$$

##### Median

Since **n = 10 (Even)**

$$Median = \frac{46+48}{2} = \frac{94}{2} = 47$$

##### Mode

45 appears **2 times**, while all other values appear only once.

Therefore,

**Mode = 45**

#### Steps

1. Find the total of all observations.

2. Divide by the number of observations to get the mean.
3. Arrange the data in ascending order.
4. Find the middle two values and calculate the median.
5. Identify the value that occurs most frequently to find the mode.

**Final Answer**

- **Mean = 51.3%**
- **Median = 47%**
- **Mode = 45%**

**Q2. Calculate the Range.**

**Answer**

**Range = 50**

**Given**

- Maximum Value = 90
- Minimum Value = 40

**Formula**

$$Range = Maximum - Minimum$$

**Calculation**

$$90 - 40 = 50$$

**Steps**

1. Identify the maximum value.
2. Identify the minimum value.
3. Subtract the minimum from the maximum.
4. Write the final answer.

**Final Answer**

**Range = 50**

**Q3. Find Q1, Q3 and IQR.**

**Answer**

- $Q1 = 45$
- $Q3 = 52$
- $IQR = 7$

#### **Calculation**

Lower Half:

40, 42, 45, 45, 46

$Q1 = 45$

Upper Half:

48, 50, 52, 55, 90

$Q3 = 52$

#### **Formula**

$$IQR = Q3 - Q1$$

#### **Calculation**

$$52 - 45 = 7$$

#### **Steps**

1. Arrange the data in ascending order.
2. Divide the data into two halves.
3. Find the median of the lower half ( $Q1$ ).
4. Find the median of the upper half ( $Q3$ ).
5. Calculate IQR.

#### **Final Answer**

- $Q1 = 45$
- $Q3 = 52$
- $IQR = 7$

#### **Q4. Identify Possible Outliers.**

#### **Answer**

**90 is a possible outlier.**

**Given**

- $Q1 = 45$
- $Q3 = 52$
- $IQR = 7$

**Formula**

Lower Limit

$$Q1 - 1.5(IQR)$$

Upper Limit

$$Q3 + 1.5(IQR)$$

**Calculation**

Lower Limit

$$45 - (1.5 \times 7) 45 - 10.5 = 34.5$$

Upper Limit

$$52 + (1.5 \times 7) 52 + 10.5 = 62.5$$

Since **90** > **62.5**, it is a possible outlier.

**Steps**

1. Find the IQR.
2. Calculate the lower limit.
3. Calculate the upper limit.
4. Compare all observations with the limits.
5. Identify the outlier.

**Final Answer**

**90 is a Possible Outlier.**

**Q5. Determine the Direction of Skewness.****Answer**

The dataset is **Right-Skewed (Positively Skewed)**.



### Comparison Table

| Observation               | Interpretation              |
|---------------------------|-----------------------------|
| Mean (51.3) > Median (47) | Right Skew                  |
| High Value (90)           | Pulls the Mean to the Right |

### Steps

1. Compare the mean and median.
2. Observe the presence of the extreme value (90).
3. Since the mean is greater than the median, the distribution is right-skewed.
4. Write the conclusion.

### Final Answer

**The CPU usage data is Right-Skewed (Positively Skewed).**

### CASE STUDY – 5: Comparing Two Datasets

#### Given Data

**Dataset A 10 20 30 40 50**

Dataset B 28 29 30 31 32

**Q1. Calculate the Mean of both datasets.**

#### Answer

- **Mean of Dataset A = 30**
- **Mean of Dataset B = 30**

#### Given

Dataset A = 10, 20, 30, 40, 50

Dataset B = 28, 29, 30, 31, 32

Number of observations (**n**) = 5

#### Formula

$$Mean = \frac{\Sigma X}{n}$$

#### Calculation

### Dataset A

$$10 + 20 + 30 + 40 + 50 = 150 \text{ Mean} = \frac{150}{5} = 30$$

### Dataset B

$$28 + 29 + 30 + 31 + 32 = 150 \text{ Mean} = \frac{150}{5} = 30$$

### Steps

1. Find the sum of all values.
2. Count the number of observations.
3. Apply the mean formula.
4. Calculate the mean.
5. Write the final answer.

### Final Answer

- **Dataset A Mean = 30**
- **Dataset B Mean = 30**

### Q2. Calculate the Variance of both datasets.

#### Answer

- **Variance of Dataset A = 200**
- **Variance of Dataset B = 2**

#### Formula

$$\text{Variance} = \frac{\sum (X - \mu)^2}{N}$$

#### Calculation

##### Dataset A (Mean = 30)

| X  | X - 30 | (X - 30) <sup>2</sup> |
|----|--------|-----------------------|
| 10 | -20    | 400                   |
| 20 | -10    | 100                   |
| 30 | 0      | 0                     |
| 40 | 10     | 100                   |

|    |    |     |
|----|----|-----|
| 50 | 20 | 400 |
|----|----|-----|

$$\sum(X - \mu)^2 = 1000 \text{ Variance} = \frac{1000}{5} = 200$$

#### Dataset B (Mean = 30)

| X  | X - 30 | (X - 30) <sup>2</sup> |
|----|--------|-----------------------|
| 28 | -2     | 4                     |
| 29 | -1     | 1                     |
| 30 | 0      | 0                     |
| 31 | 1      | 1                     |
| 32 | 2      | 4                     |

$$\sum(X - \mu)^2 = 10 \text{ Variance} = \frac{10}{5} = 2$$

#### Steps

1. Find the mean.
2. Calculate each deviation.
3. Square the deviations.
4. Find the total.
5. Divide by N.

#### Final Answer

- Dataset A Variance = 200
- Dataset B Variance = 2

**Q3. Calculate the Standard Deviation of both datasets.**

#### Answer

- Dataset A Standard Deviation = 14.14

- **Dataset B Standard Deviation = 1.41**

#### **Formula**

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

#### **Calculation**

##### **Dataset A**

$$SD = \sqrt{200} = 14.14$$

##### **Dataset B**

$$SD = \sqrt{2} = 1.41$$

#### **Steps**

1. Find the variance.
2. Apply the standard deviation formula.
3. Calculate the square root.
4. Write the final answer.

#### **Final Answer**

- **Dataset A SD = 14.14**
- **Dataset B SD = 1.41**

#### **Q4. Which dataset has greater spread?**

#### **Answer**

**Dataset A has greater spread** because it has a higher variance and standard deviation.

#### **Comparison Table**

##### **Dataset    Variance    Standard Deviation    Spread**

|               |       |  |         |
|---------------|-------|--|---------|
| Dataset A 200 | 14.14 |  | Greater |
| Dataset B 2   | 1.41  |  | Smaller |

#### **Steps**

1. Compare the variances.
2. Compare the standard deviations.
3. The dataset with the higher values has greater spread.

4. Write the conclusion.

### Final Answer

**Dataset A has greater spread.**

**Q5. If both datasets represent model errors, which model appears more consistent?**

### Answer

**Dataset B appears more consistent** because it has a much lower variance and standard deviation.

### Comparison Table

#### Dataset Consistency

Dataset A Less Consistent

Dataset B More Consistent

### Steps

1. Compare the standard deviations.
2. Lower standard deviation indicates higher consistency.
3. Dataset B has the lower standard deviation.
4. Therefore, Dataset B is more consistent.

### Final Answer

**Dataset B is more consistent.**

**Q6. Explain why mean alone is not enough to compare datasets.**

### Answer

The **mean alone is not enough** because two datasets can have the same mean but different spreads. Variance and standard deviation show how the values are distributed around the mean, making comparison more accurate.

### Comparison Table

| Measure | Dataset A | Dataset B |
|---------|-----------|-----------|
| Mean    | 30        | 30        |

|                    |       |      |
|--------------------|-------|------|
| Variance           | 200   | 2    |
| Standard Deviation | 14.14 | 1.41 |

### Steps

1. Compare the means.
2. Observe that both means are equal.
3. Compare the variance and standard deviation.
4. Conclude that spread is also important.

### Final Answer

**Mean alone cannot compare datasets completely because it does not show the variation in data. Variance and standard deviation must also be considered.**